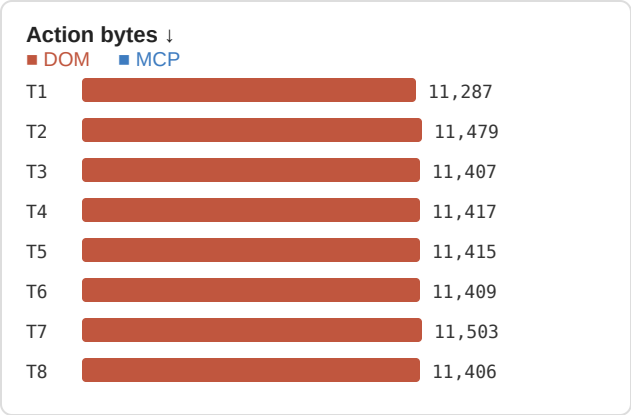
WebMCP Real-Browser Comparison — Run 2

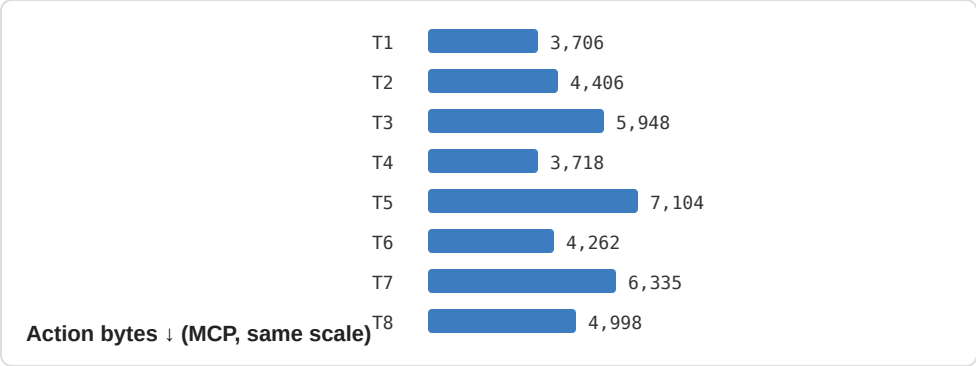
Generated 13 Oct 2026 07:34 UTC · brain: glm-5.3-flash (temp 0) · 8 shopping tasks × 2 sites · all site traffic inside headless Chrome; identical stores, only the MCP layer differs

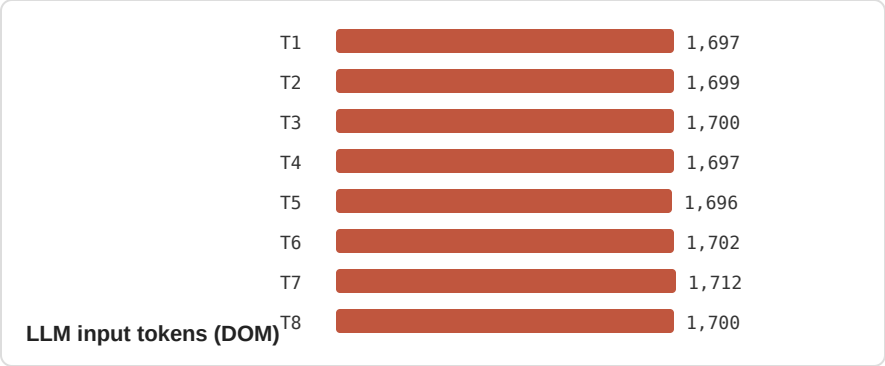
Headline (Run 2)

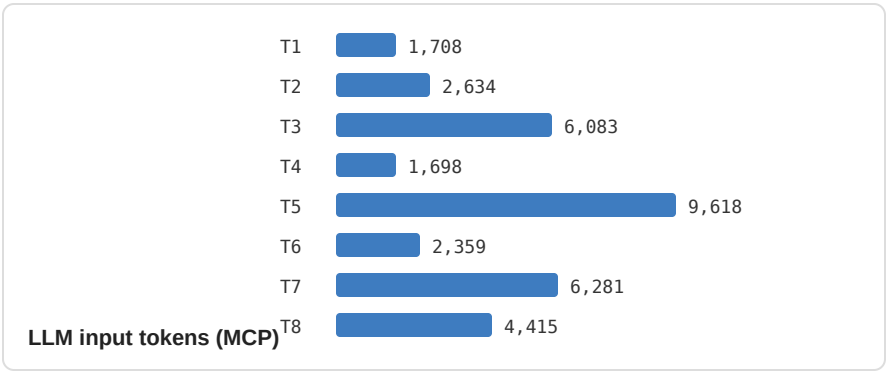
	Success	Action data ↓	Requests	LLM tok in	LLM tok out	Agent time	Wire page bytes
DOM (demonomcp)	**8/8**	91,323 B	32	13,603	2,461	25.3 s	154,835 B
MCP (demowithmcp)	**8/8**	**40,477 B (−56%)**	55	34,796	3,659	74.3 s	144,729 B

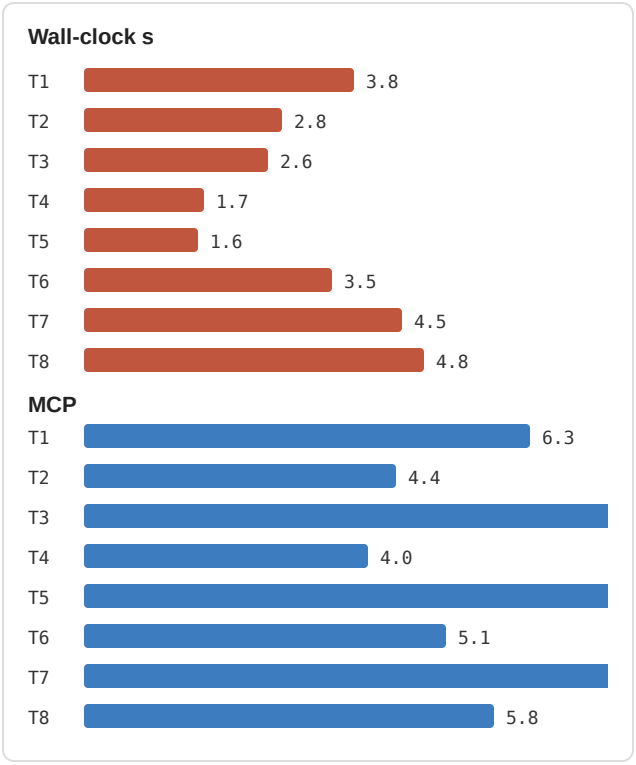
Task-by-task (Run 2) — DOM vs MCP











Choices & orders placed (Run 2)

Task	DOM pick	Order	MCP pick	Order	Same choice?
T1	Logitech M235 Wireless	—	Logitech M235 Wireless	—	✓ identical
T2	HP Pavilion 14	1007	HP Pavilion 14	—	✓ identical
T3	iQOO Z9s Pro	1008	iQOO Z9s Pro	1006	✓ identical
T4	Razer DeathAdder Essential	1009	Razer DeathAdder Essential	—	✓ identical
T5	MacBook Air M3 13	1010	MacBook Air M3 13	1007	✓ identical
T6	Realme 13 Pro+	1011	Realme 13 Pro+	1008	✓ identical
T7	Logitech MX Master 3S	1012	Logitech MX Master 3S	1009	✓ identical
T8	OnePlus 12R	1013	OnePlus 12R	1010	✓ identical

Run-to-run stability (Run 1 vs Run 2, real browser)

	Run 1 (07:18 UTC)	Run 2
DOM — success	8/8	8/8
MCP — success	8/8	8/8
DOM — action bytes	91,266	91,323 (+0.06%)
MCP — action bytes	41,753	40,477 (−3.1%)

DOM — LLM tok in	13,603	13,603 (identical)
MCP — LLM tok in	39,495	34,796 (−11.9%)
Choices identical run1↔run2	**✓ all 16 picks match (deterministic at temp 0)**	

Context: simulated vs real browser

	Simulated (curl, 12 Oct)		Real browser	
	DOM	**MCP**	**DOM**	**MCP**
Action bytes	101,465	64,723	91,323	40,477
Requests	40	87	32	55
LLM tok in	13,635	23,187	13,603	34,796
Success	**8/8**	**8/8**	**8/8**	**8/8**

Findings

- **MCP cuts per-task site data ~56%** in real-browser execution (40KB vs 91KB over 8 tasks) — structured tool responses instead of dragging the whole catalog through the context.
- **DOM path is cheaper for the LLM**: one catalog read covers the task (1–2 LLM calls); MCP needs 2–4 calls and re-sends tool schemas each turn — ~2.6× input tokens.
- **DOM path is faster** (25s vs 74s) — fewer round-trips. Both complete all tasks correctly.
- **Deterministic**: all 16 picks identical across both runs at temperature 0; action bytes vary ≤3%.
- Trade-off: MCP = context economy on the wire; DOM = fewer turns, cheaper tokens, faster. On a catalog this small DOM wins on cost/speed; MCP wins as data volume grows.

Artifacts: /root/webmcp_demo/browser_llm.py · results_browser_run1.json · results_browser_run2.json · sites: demowithmcp / demonomcp.converso.work